ASEAN
GEOAI FUSION 2026

TRAINING MODULE

Fundamentals of Machine Learning

Core concepts, primary tasks, the AI/ML/DL stack, and frameworks



What you'll be able to do

- **Describe** core fundamentals of machine learning and how models learn from data.
- **Differentiate** AI, Machine Learning, and Deep Learning as nested layers.
- **Identify** the three primary ML tasks and when to apply each.
- **Use** a machine learning framework effectively to build and deploy models.

01	02	03	04
Core ML fundamentals	AI vs ML vs DL	Primary tasks	Frameworks

How the module is structured

- **1.0** Background — How the machine learns and the three learning types.
- **2.0** AI vs ML vs Deep Learning — the nested technology stack.
- **3.0** Primary ML tasks — Classification, Regression, Clustering.
- **4.0** Machine Learning Frameworks — tools to build and deploy models.
- **5.0** Summary — key takeaways and next steps.

Module path

You are here → 1.0 Introduction

SECTION 1.0

Background

How the machine learns, and the three ways machines learn from data.

How the machine learns?

ML is a subset of artificial intelligence that enables computers to:

- **Learn** from data — patterns, relationships, signals.
- **Make predictions** — about new, unseen instances.
- **Improve** automatically as more data becomes available.

Why it matters

Machines can find structure in data that humans cannot — at scale, at speed, and without being explicitly programmed for every scenario.

FOUNDATION

Understanding how machines learn requires a grasp of:

- How data is processed
- How models evaluate performance
- How learning approaches are categorized

Three ways machines learn

Learning approaches are categorized by the kind of supervision available:

01 Supervised

Trains on labeled data — both input and expected output are known.

TYPICAL USE

Used for predicting values or categories.

02 Unsupervised

Evaluates unlabeled data to find hidden structures, clusters, or anomalies.

TYPICAL USE

Used for discovery and segmentation.

03 Reinforcement

Learns through trial and error — actions are adjusted by a system of rewards and penalties.

TYPICAL USE

Used for sequential decision-making.

SECTION 2.0

AI vs ML vs Deep Learning

These three terms are not synonyms — they are nested layers of the same stack.

The nested technology stack

ARTIFICIAL INTELLIGENCE (AI) — any technique that mimics human intelligence

MACHINE LEARNING (ML) — systems that learn from data without explicit programming

DEEP LEARNING (DL)

ML that uses brain-inspired neural networks with multiple layers

auto-discovers the features needed for the task

Not all AI is ML. All ML is AI. Deep learning is a specialized branch of ML.

Real-world examples by layer

AI — the universe

- Siri: voice command understanding
- ChatGPT: human-like text responses
- Self-driving cars: traffic navigation
- Game-playing AI: chess, Go champions
- Recommendation systems: Netflix suggestions
- Spam filters: deciding if email is junk

ML — the planet

- Spam filters learn from email patterns
- Amazon product recommendations
- Credit card fraud detection
- Predictive text on your phone
- Medical disease diagnosis from imaging

Deep Learning — the city

- Face recognition (phone unlock)
- Large language models: ChatGPT, Claude
- Image generation: DALL·E, Midjourney
- Natural-sounding voice synthesis
- Autonomous vehicle perception
- Medical imaging: tumor detection in CT scans

SECTION 3.0

Primary ML Tasks

Three foundational tasks — Classification, Regression, and Clustering.

Classification · Regression · Clustering

Classification

Predicting categorical variables

"Which category does this belong to?"

Supervised · Discrete categories

EXAMPLE

e.g. classifying an email as spam or not spam

Regression

Predicting continuous values

"How much or how many?"

Supervised · Continuous numbers

EXAMPLE

e.g. predicting house prices

Clustering

Grouping similar data points

"How do these naturally group?"

Unsupervised · Group memberships

EXAMPLE

e.g. customer segmentation

Predicting a discrete categorical label

A supervised task that maps inputs to one of N categories.

COMMON ALGORITHMS

- Logistic Regression
- Decision Trees and Random Forests
- Support Vector Machines (SVM)
- K-Nearest Neighbors (KNN)
- Naive Bayes
- Neural Networks

VARIANTS

- **Binary** — exactly two classes (spam / not spam).
- **Multiclass** — three or more exclusive classes.
- **Multi-label** — multiple non-exclusive tags per item.

REAL-WORLD EXAMPLES

- Email filtering — spam vs. inbox
- Medical diagnosis — malignant vs. benign
- Credit card fraud detection
- Sentiment analysis — pos / neg / neutral
- Image recognition — cat, dog, bird

EVALUATION

Accuracy, Precision, Recall, F1, ROC-AUC

Predicting a continuous numerical value

A supervised task that outputs a real number on a continuous scale.

COMMON ALGORITHMS

- Linear Regression
- Polynomial Regression
- Ridge and Lasso Regression
- Decision Tree Regression
- Random Forest Regression
- Gradient Boosting (XGBoost, LightGBM)
- Neural Networks

VARIANTS

- **Linear** — straight-line relationship.
- **Polynomial** — curved / non-linear fits.
- **Multivariate** — many features, one output.
- **Time-series** — forecast future values.

REAL-WORLD EXAMPLES

- House prices — size, location, amenities
- Stock prices or sales revenue forecasts
- Weather estimates — temperature, rainfall
- Customer lifetime value
- Insurance risk and premium pricing

EVALUATION

MAE, MSE, RMSE, R-squared (R^2)

Grouping similar data without labels

An unsupervised task that discovers structure in unlabeled data.

COMMON ALGORITHMS

- K-Means Clustering
- Hierarchical Clustering (Agglomerative / Divisive)
- DBSCAN — Density-Based Spatial Clustering
- Mean Shift
- Gaussian Mixture Models (GMM)
- Spectral Clustering

TYPES

- **Centroid-based** (K-Means).
- **Density-based** (DBSCAN).
- **Hierarchical** (dendrograms).
- **Distribution-based** (GMM).

REAL-WORLD EXAMPLES

- Customer segmentation for marketing
- Document and news article grouping
- Anomaly detection — outliers form clusters
- Gene sequence analysis (bioinformatics)
- Image compression and color quantization

EVALUATION

Silhouette, Davies-Bouldin, Inertia

Comparison of primary ML tasks

Aspect	Classification	Regression	Clustering
Learning type	Supervised	Supervised	Unsupervised
Output type	Discrete categories	Continuous numbers	Group memberships
Requires labels	Yes	Yes	No
Goal	Predict a class	Predict a value	Discover structure
Key challenge	Class imbalance	Overfitting, outliers	Choosing K (clusters)
Example	Spam detection	House price prediction	Customer segmentation

Choose the right task based on your data, your objective, and whether labels are available.

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THANK YOU

Bootcamp and Hackathon for ASEAN Geospatial Innovation 2026